Probabilistic Robotics Exam Sample
Dec 12 2019

January 7, 2020

1 Question: Least Squares

Derive the Gauss-Newton algorithm, and motivate the fact that the coefficient matrix $\mathbf{H}$ at the equilibrium is the inverse covariance of the estimate.

1.1 Solution

See slides 18 (from 2 to 9) and slides 21 (from 6 to 12) for the complete derivation

Hints :

- Introduce the generic problem:
  - We want to estimate a generic set of parameters $\mathbf{x} \in \mathbb{R}^n$
  - We have available a set of indirect observations about the parameters, $\mathbf{z}^{[i]} \in \mathbb{R}^m$, $i \in \{1 \dots N\}$
  - We know a (non-linear) prediction function that, given the actual value of the parameters, can infer the value of the observation i, $\mathbf{h}^{[i]}(\mathbf{x})$
  - We define the error function as :

$$\mathbf{e}^{[i]}(\mathbf{x}) = \mathbf{h}^{[i]}(\mathbf{x}) - \mathbf{z}^{[i]}$$

  - The problem can be then formulated as:

$$\mathbf{x}^* = \underset{\mathbf{x}}{\operatorname{argmin}} \sum_{i=1}^{N} \|\mathbf{e}^{[i]}(\mathbf{x})\|^2_\Omega = \underset{\mathbf{x}}{\operatorname{argmin}} \sum_{i=1}^{N} \mathbf{e}^{[i]}(\mathbf{x})\, \mathbf{\Omega}\, \mathbf{e}^{[i]}(\mathbf{x})$$

    Where $\mathbf{\Omega}$ is either the inverse covariance of the observations (if you use the probabilistic interpretation, slides 18 from 2 to 5), or a matrix of weights which scales the different components of the error.
  - Once the problem is defined you can follow the derivation on the slides

2 Exercise: Bouncing Ball

Scenario: An easily detectable red ball bounces on an unlimited field, observed by a fixed pinhole camera. An algorithm processes the camera output, and returns the center and the radius of the ball in image coordinates. The radius r of the ball is known, and the ball motion is subject to air fricton $\alpha < 1$. Once it bounces against the floor (which is the $x - y$ plane), the velocity component along the z axis changes sign. In short, the motion of the ball is approximated by the following equation:

$$\begin{pmatrix} v_{x,t+1} \\ v_{y,t+1} \\ v_{z,t+1} \end{pmatrix} = \begin{cases} \alpha \begin{pmatrix} v_{x,t} \\ v_{y,t} \\ v_{z,t} \end{pmatrix} + \mathbf{g} \cdot \Delta T & \text{if } z > r \\ \alpha \begin{pmatrix} v_{x,t} \\ v_{y,t} \\ \|v_{z,t}\| \end{pmatrix} + \mathbf{g} \cdot \Delta T & \text{otherwise} \end{cases} \tag{1}$$

Here ΔT is the (small!) time interval of the discrete system.
The velocity is subject to a small additive gaussian noise $\mathbf{n}_v \sim \mathcal{N}(\mathbf{0}, \Sigma_v)$.

2.1 Question 1

Identify the system by defining:

- State
- Controls (if any)
- Transition function
- Measurement function assuming the camera is at position $\mathbf{T}_{\text{cam}}$, and has known camera parameters $\mathbf{K}$. The camera has *square* pixels ($f_x = f_y$).

Hint: assume the ball is far enough from the camera and its observed radius depends on the inverse of the depth.

2.2 Question 2

After having identified the system, approach the problem of estimating the ball position with a filter at your choice. Report the filter equations, for the specific problem and a sketch of filter implementation. If Jacobians are involved report the sizes.

2.3 Question 3

Discuss the pro and the contra of PF, KF and UKF in this particular problem.

2.4 Solution : Question 1

- State : $\mathbf{x} = \{\mathbf{p}, \mathbf{v}\}$, where $\mathbf{p}$ is the position of the ball center in world frame, and $\mathbf{v}$ it's velocity.
- No controls, the ball is in free motion.
- The transition for $\mathbf{v}$ is given in (1). The one for the position can be obtained by Euler integration, as follows:

$$\begin{pmatrix} p_{x,t+1} \\ p_{y,t+1} \\ p_{z,t+1} \end{pmatrix} = \begin{cases} \begin{pmatrix} p_{x,t} \\ p_{y,t} \\ p_{z,t} \end{pmatrix} + \alpha \begin{pmatrix} v_{x,t} \\ v_{y,t} \\ v_{z,t} \end{pmatrix} \Delta T + \mathbf{g} \cdot \Delta T^2 & \text{if } z > r \\ \begin{pmatrix} p_{x,t} \\ p_{y,t} \\ p_{z,t} \end{pmatrix} + \alpha \begin{pmatrix} v_{x,t} \\ v_{y,t} \\ \|v_{z,t}\| \end{pmatrix} \Delta T + \mathbf{g} \cdot \Delta T^2 & \text{otherwise} \end{cases}$$

- The measurement function for the ball center is a standard projective model, with camera at $\mathbf{T}_{cam}$, on a single point
$$\hat{\mathbf{z}}_c(\mathbf{x}) = \mathrm{proj}(\mathbf{K}\,\mathbf{T}_{cam}^{-1}\,\mathbf{p}_t)$$

- The ball will appear in the image as a disk, and the points on the disk describe a contour on the ball surface in 3D. If the ball is sufficiently far from the observer, we can assume that this contour is obtained by the intersection of a plane passing through the ball center and parallel to the image plane. If the camera is at the origin, and the ball center is at $\mathbf{p}$, the points on of such a disk, $\mathbf{d}(\theta)$ in *camera* coordinates are
$$\mathbf{d}(\theta) = \mathbf{p} + \begin{pmatrix} r \cdot \cos(\theta) \\ r \cdot \sin(\theta) \\ 0 \end{pmatrix} \tag{2}$$
These points will be imaged as
$$\mathbf{b}(\theta) = \begin{pmatrix} \frac{f(r \cdot \cos(\theta) + p_x)}{p_z} + c_x \\ \frac{f(r \cdot \sin(\theta) + p_y)}{p_z} + c_y \end{pmatrix}, \tag{3}$$
while the center will be imaged as:
$$\mathbf{c} = \begin{pmatrix} \frac{f \cdot p_x}{p_z} + c_x \\ \frac{f \cdot p_y}{p_z} + c_y \end{pmatrix}. \tag{4}$$
Here $\mathbf{p} = (p_x, p_y, p_z)^T$ is the center of the ball, f is the focal length and c_x, c_y is the principal point on the image. The measured radius is just the difference between the center of projection (4) and any point on the border (3):
$$\|\mathbf{c} - \mathbf{b}(\theta)\| = \left\| \begin{matrix} \frac{f \cdot r \cdot \cos(\theta)}{p_z} \\ \frac{f \cdot r \cdot \sin(\theta)}{p_z} \end{matrix} \right\| = \frac{f \cdot r}{p_z}. \tag{5}$$
Which represent how the radius itself is projected through a pinhole camera model. Further we know that the radius (*as any distance*) is invariant with respect to rigid transformations. As such, given the actual estimate of the ball center and the known ball radius, our measurement function can be written as :
$$\mathbf{p}'_t = \mathbf{T}_{cam}^{-1} \cdot \mathbf{p}_t \tag{6}$$
$$\hat{\mathbf{z}}_r(\mathbf{x}) = \frac{f \cdot r}{p'_{z,t}}. \tag{7}$$
Here $\mathbf{p}' = (p'_x, p'_y, p'_z)^T$ is the z coordinate of the ball in the reference frame of the camera.

The complete measurement function is then :
$$\hat{\mathbf{z}}(\mathbf{x}) = \begin{pmatrix} \hat{\mathbf{z}}_c(\mathbf{x}) \\ \hat{\mathbf{z}}_r(\mathbf{x}) \end{pmatrix} \tag{8}$$

2.5 Solution Question 2

2.6 EKF

Question 2 : In this case any filter is a valid answer. Mostly try to highlight pros (if any), contra (if any) and motivation of your choice. In this section, I will just highlight the limitations of the EKF for this particular problem.
The transition model is discontinuous and non-linear, due to both the approximation underlying the euler integration and the presence of $\|v_{z,t}\|$ for $z < r$. If we further consider that the dimension of the state is rather small (6), the EKF seems the less suitable filtering approach for this problem (you can rather use an UKF or PF).

2.6.1 Jacobians

- Jacobian of the transition model
$$\mathbf{A}_t \in \mathbb{R}^{6\times 6}$$
- Jacobian of the measurement function
$$\mathbf{C}_t \in \mathbb{R}^{3\times 6}$$

2.6.2 Filter Equations

Initial State :

$$p(\mathbf{x}_{0,0}) \sim \mathcal{N}(\mathbf{x}; \mu_{0,0}, \Sigma_{0,0})$$

Transition :

$$\begin{aligned}
\mu_{t|t-1} &= \mathbf{A}_t \mu_{t-1|t-1} \\
\mathbf{\Sigma}_{t|t-1} &= \mathbf{A}_t \mathbf{\Sigma}_{t-1|t-1} \mathbf{A}_t^T + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \Sigma_v \end{bmatrix}
\end{aligned}$$

Correction :

$$\begin{aligned}
\mu_z &= \mathbf{C}_t \mu_{t|t-1} \\
\mathbf{K}_t &= \mathbf{\Sigma}_{t|t-1} \mathbf{C}_t^T \left(\mathbf{\Sigma}_z + \mathbf{C}_t \mathbf{\Sigma}_{t|t-1} \mathbf{C}_t^T \right)^{-1} \\
\mu_{t|t} &= \mu_{t|t-1} + \mathbf{K}_t \left(\mathbf{z}_t - \mu_z \right) \\
\mathbf{\Sigma}_{t|t} &= \left(\mathbf{I} - \mathbf{K}_t \mathbf{C}_t \right) \mathbf{\Sigma}_{t|t-1}
\end{aligned}$$

2.7 Solution Question 3

General pro/contra of the filtering approches (in short) :

- EKF :
  - Cannot properly handle highly non-linear systems (**contra**)
  - If the state has dimension n we need to store only n numbers (**pro**).
  - We need to compute the jacobians (**contra**)
  - The state is assumed to be distributed like a gaussian.
- UKF :
  - Can handle non-linearities (**pro**)
  - If the state has dimension n we need to store $n \times (2n+1)$ numbers (**contra**).
  - No jacobians (**pro**)
  - The state is assumed to be distributed like a gaussian.
- PF :
  - Can handle non-linearities (**pro**)
  - If the state has dimension n and we use M particles we need to store $n \times M$ numbers (**contra**).
  - The convergence of the filter is related to the number of particles (the more the better). (**contra** if combined with the previous point)
  - No jacobians (**pro**)
  - The state can be arbitrary distributed

3 Exercise: Fit The Circle

Scenario: Given a set of points on a plane, we want find a subset of these points that belong a single circle of unknown radius, and center.

3.1 Question 1

Approach the problem with RANSAC by

- Defining the parameters to be fitted
- Defining a minimal solver
- Defining an error function to determine inliers and outliers

3.2 Question 2

Given a set of inliers and an initial estimate, formulate the least squares problem to optimally refine the parameters by considering only the inliers.

3.3 Question 3

Assuming that only 10% of the points belong to the circle, how many rounds of RANSAC are needed to compute a solution with probability $p = 0.99$?

4 Solution Question 1

4.1 Parameters

A minimal representation of a circle is given by a 2D point $\mathbf{c}$ (center) and a scalar r (radius).

4.2 Minimal Solver

- Pick three random points and for each of them construct the equation :

$$\|\mathbf{p}^{[i]} - \mathbf{c}\|^2 - r^2 = \|\mathbf{c}\|^2 + \|\mathbf{p}^{[i]}\|^2 - 2 \cdot \mathbf{c}^T \cdot \mathbf{p}^{[i]} - r^2 = 0 \tag{9}$$

- Construct the system by stacking the three equations :

$$\begin{cases} \|\mathbf{c}\|^2 + \|\mathbf{p}^{[1]}\|^2 - 2 \cdot \mathbf{c}^T \cdot \mathbf{p}^{[1]} - r^2 = 0 \\ \|\mathbf{c}\|^2 + \|\mathbf{p}^{[2]}\|^2 - 2 \cdot \mathbf{c}^T \cdot \mathbf{p}^{[2]} - r^2 = 0 \\ \|\mathbf{c}\|^2 + \|\mathbf{p}^{[3]}\|^2 - 2 \cdot \mathbf{c}^T \cdot \mathbf{p}^{[3]} - r^2 = 0 \end{cases} \tag{10}$$

- Subtract the first from the other two and create the two linear equations :

$$\begin{cases} 2 \cdot \mathbf{c}^T \cdot (\mathbf{p}^{[2]} - \mathbf{p}^{[1]}) = \|\mathbf{p}^{[1]}\|^2 - \|\mathbf{p}^{[2]}\|^2 \\ 2 \cdot \mathbf{c}^T \cdot (\mathbf{p}^{[3]} - \mathbf{p}^{[1]}) = \|\mathbf{p}^{[1]}\|^2 - \|\mathbf{p}^{[3]}\|^2 \end{cases} \tag{11}$$

  Then solve for $\mathbf{c}$.

- Plug the solution of $\mathbf{c}$ in the first equation of (10) and solve for r.

practical consideration : To enhance robustness to noise, you can impose a minimum distance between the three random points.

4.3 Error function

Given the current value of $(\mathbf{c}, r)$ and a query point $\mathbf{p}$, a possible error function is:

$$e(\mathbf{c}, r, \mathbf{p}) = \|\mathbf{p} - \mathbf{c}\| - r \tag{12}$$

Given a threshold t (user defined) a point is considered an inlier if:

$$|e(\mathbf{c}, r, \mathbf{p})| < t \tag{13}$$

5 Solution Question 2

- State : $\mathbf{x} = (\mathbf{c}, r)^T \in \mathbb{R}^3$ (Euclidean)
- Error Function : same as in equation (12)
- Jacobian : $\mathbf{J} \in \mathbb{R}^{1\times 3}$

In this case we don't have celarly identifiable measurements since a sensor model is not provided. Still we can define an error function by looking at the model/parameters that we are estimating. In this case, our error function evaluates how much each point is consistent with the current estimate of the circle.

6 Solution Question 3

RANSAC (slides 26 slide number 14)
Apply the formula:

$$k = \frac{\log(1-p)}{\log(1-w^n)} \tag{14}$$

With $p = 0.99$ (probability of success), $w = 0.1$ (inlier ratio), $n = 3$ (number of data points required by the model (*aka minimal solver*)).